

Anh (Andre) Tran

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EDUCATION

University of Cincinnati

Expected May 2028

Bachelor of Science in Computer Engineering, Minor in Computer Science

GPA: 3.77/4.00

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Design & Development, Cybersecurity, Digital Design & Digital Systems, Artificial Intelligence, Context Engineering

EXPERIENCE

University of Cincinnati Department of Computer Science

08/2026 - Present

Incoming AI Researcher

Cincinnati, OH

- Building AI-driven voice modification and exercise-recommendation modules (e.g., PerMod) in Dr. Novak's lab for **TruVox**, an existing gender-affirming voice training platform, alongside its deployment infrastructure.

Commma - Pace your Code (commma.dev)

05/2026 - Present

Founder & Lead Engineer

Cincinnati, OH

- Building **commma.dev** end-to-end, a **TypeScript** monorepo spanning a **VS Code/Open VSX** extension, a **Hono/Postgres** API, and a React SPA, live on AWS at 100% availability over 30 days (500+ downloads).
- Replaced **BullMQ/Redis** with an interval scanner, holding Redis commands near 0 against a 500K/mo budget cap.
- Enforced a keys-only privacy guarantee at two layers: the extension never reads editor content, and the API re-derives privacy mode from a **Redis**-cached lookup, cutting p50 latency 42ms to 13ms.
- Load-tested the live stack to 271K events / 4,447 sessions with zero 5xx, then traced a local connection-pool win that didn't reproduce in prod to client bandwidth, not the server.

Caphne - Study Buddies Matchmaking (caphne.co)

01/2026 - 08/2026

Lead Software Engineer

Remote

- Cut API response time from 60s to 10s by engineering a distributed, real-time chat system with **Socket.IO** scoped to per-match rooms, sustaining 1.7K+ requests/minute over 30 days in production.
- Led a cross-functional team of 6 engineers through sprint planning, code review, and production deploys of a full-stack **Nuxt 3**, **Vue**, and **Node.js** platform, scaling to 400+ active users.
- Architected **AWS** infrastructure as code (EC2, CloudFront, S3, Route 53) via **Terraform** with **OIDC**-federated credentials, then root-caused and fixed 2 production incidents same-day from raw error output.
- Deleted two speculative database tables from match rate-limiting, raising daily match supply 7x (3/3hr to 20/day).

PROJECTS

Kiln-IDP (Live) | *Go, Java (Spring Boot), Kubernetes, ArgoCD, Istio, Kafka, PostgreSQL, OpenTelemetry*

- Cut cluster spend 45.5% with zero SLA violations via a custom **Kubernetes** scheduler plugin.
- Designed a canary controller around a Wald sequential probability ratio test with a cumulative CUSUM across the rollout, catching 100% of 1,000 seeded regressions at 2.8% false-rollback rate on a live **Istio** mesh.
- Replaced ticket-and-Terraform provisioning (hours to a day) with a self-service path reaching Ready at 11s p50, integrating 6 subsystems behind one fail-closed policy gate.

Draft-Thinker (Live) | *Go, OpenAI API, Qdrant, Redis, Prometheus, Grafana, Docker*

- Cut inference cost 82.9% (bootstrapped 95% CI: 79.6-86.0%) while holding accuracy at 98.1% by designing a machine learning entropy router in **Go** using top-k logprobs from the **OpenAI API**.
- Diagnosed why a confidence-threshold baseline plateaus at a 61.5% cost-reduction ceiling from an OpenAI logprob-reporting artifact affecting 65.5% of creative prompts, which entropy routing structurally avoids.
- Built a speculative-execution pipeline with **Go** goroutines, wasting only 5-10% compute on 30% of requests.

Ferrox (Live) | *Rust, Systems Programming, Atomics, Mmap, Criterion*

- Built a low-latency, price-time matching engine in **Rust** with fixed-point prices, trading a 9-22% insert regression for a 67-85% cut in cancel latency via a HashMap-to-BTreeMap swap, benchmarked at 500ns P99 across 4.7M orders/sec.
- Engineered a lock-free SPSC ring buffer with cache-line-padded atomics, choosing Acquire/Release over SeqCst to avoid unnecessary fence instructions, measuring an 8.8x throughput gain over **std::sync::mpsc**.
- Deployed a containerized service to **AWS Fargate**, measuring the **NLB** at 70% of total hourly cost.

TECHNICAL SKILLS

Languages: Go, JavaScript/TypeScript, SQL, Python, Rust, C/C++, C#, Java, HTML/CSS

Technologies: Nuxt.js, Vue.js, Next.js, React, Node.js, Express.js, Spring Boot, Hono, Kubernetes, Kafka, Git, Redis, LangChain/LangGraph, AWS (Amazon Web Services), Docker, PostgreSQL, MongoDB, MySQL, FastAPI, Django, ChromaDB, scikit-learn, NumPy, Socket.IO, GitHub, Grafana, Prometheus, Terraform, CI/CD, Agile/Scrum, Linux